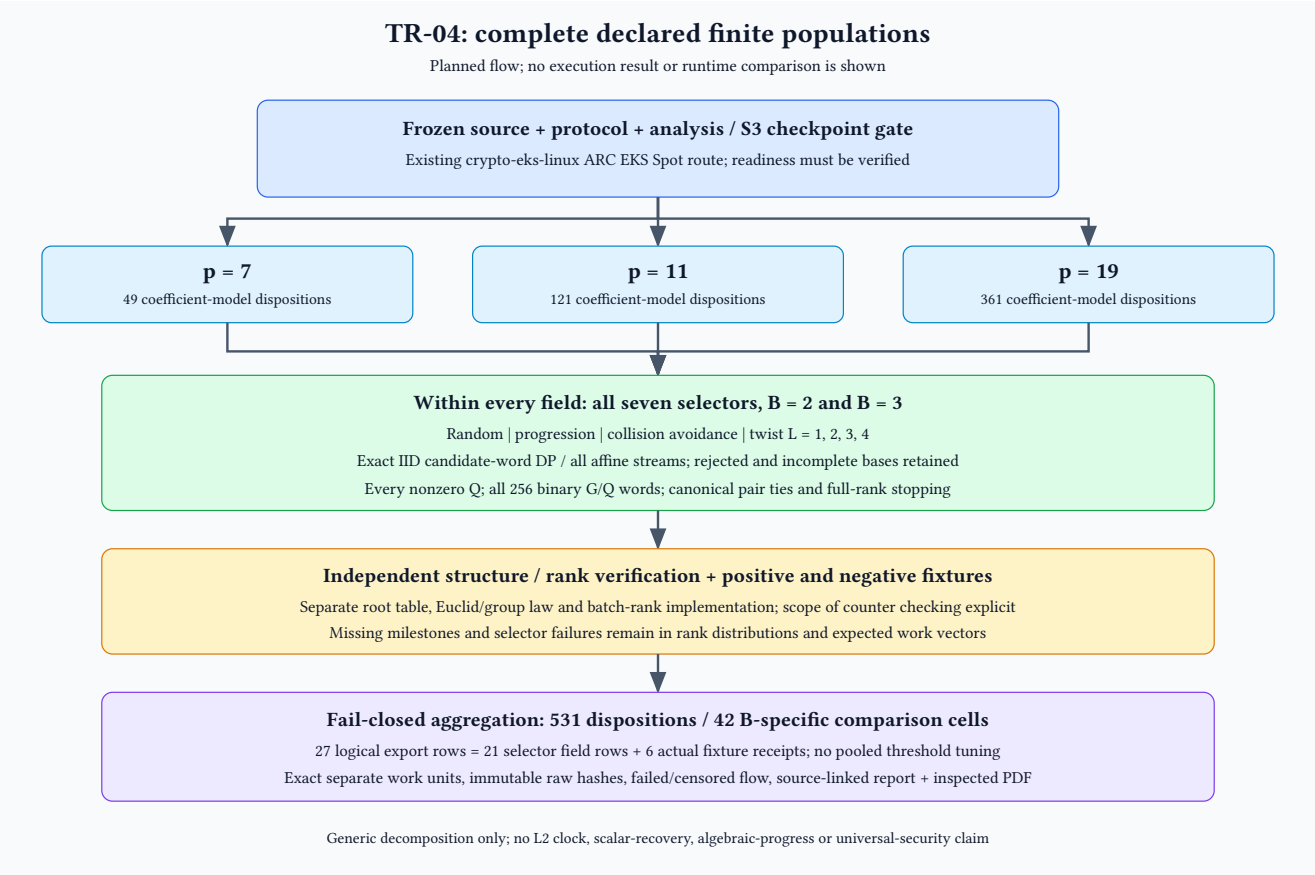
1. TR-04: exhaustive bounded relation finding

Pending-execution report source — no research result is reported here.

The user requested four steps: a bounded twist-guided selector, matched ordinary controls, an identical public original-curve relation finder, and paid work to independent relation rank. The contract in `protocol.json` and `PROTOCOL.md` was frozen at 2026-10-29 22:18:02 UTC against code `445d84bf7`, before primary comparison data. Root-library validation and publication are pending; no primary census has started.

1.1. Declared finite scope

Fields 7, 11 and 19; every coefficient pair, with 531 model dispositions. Execute eligible nonsingular prime-order models with order at least seven. Base sizes are two and three, selector budgets eight, twist thresholds one through four, and relation-word length eight. Every IID candidate word is represented exactly by a weighted subset-state DP; progressions cover every start and nonzero step. Every nonzero public target and all 256 binary G/Q words are included for the canonical generator. This is a finite census, not a universal statement about twist constructions.



Planned workflow and evidence flow. This diagram contains no measured values; actual runner, checkpoint and execution readiness remains unverified until the source-pinned preflight succeeds.

1.2. Outcomes required before this report can be completed

The run report must name the immutable source and binary hashes, protocol hash, workflow and physical job IDs, runner/host facts, complete model and stream census, six actual fixture-control receipts, independent verification coverage and every failure or missing shard. Its canonical export has 27 logical rows: nine arms by three field instances. Keep B2 and B3 separate inside the 21 selector rows, giving 42 comparison cells.

The primary quantities are exact unconditional coefficient-rank distributions and separate paid-work counter vectors. No clock is primary. DP/trie elapsed time, allocator/cache/RNG internals and unmeasured physical costs cannot be silently interpreted as a complete runtime or GAE price. A Pareto statement is valid only for its explicitly named finite model/base-size cell.

1.3. Required final visual and evidence review

After actual records arrive, replace this pending-execution narrative with observed values and the complete flow table. Add a quantitative graph from the recorded exact values, with field/base-size/threshold populations and unit labels; preserve negative and mixed findings. Record uncertainties as exact finite values rather than invented sampling

intervals. Embed that graph in the final PDF, inspect every page, and retain editable sources. If only an infrastructure failure exists, report it with this execution diagram and no fabricated comparison.

1.4. Source and scope references

SEC 1 v2, section 2.2.1: https://www.secg.org/sec1-v2.pdf. Diem, On the discrete logarithm problem in elliptic curves, section 3: https://doi.org/10.1112/S0010437X10005075. The primary Semaev ePrint PDF returned HTTP 403 during source review. The native package, PROTOCOL.md and AGENTS.md define the computational and reporting boundaries. This generic pair-table study is bounded by square-root generic complexity and makes no algebraic-track, scalar-recovery, fastest-tool, matched-rho, L2-admitted timing or general-security claim.